

CHAPTER 9

STRESS AND STRAIN

STRESS is the internal resistance set up in a material when an external force is applied.

When the applied force tends to shorten the material, or crush it, the material is said to be *in compression*, and the stress is referred to as a *compressive stress*. When the force tends to lengthen the material, or tear it apart, the material is said to be *in tension* and the stress is referred to as a *tensile stress*. When the force tends to cause the particles of the material to slide over each other, the material is said to be *in shear* and the stress is referred to as a *shear stress*.

Stress is always expressed as stress intensity and is therefore the force per unit area of the material.

$$\text{Stress} = \frac{\text{Total force}}{\text{Area}} \quad \text{N/m}^2$$

$$\sigma = \frac{F}{A} \quad \text{tens. or comp.}$$

$$\tau = \frac{F}{A} \quad \text{shear}$$

In the case of direct tensile or compressive forces, the area carrying the force is the cross-section in the plane of the material normal (i.e. perpendicular) to the direction of the force. In the case of a shear force, the area carrying the force is the area to be sheared through in the direction of the line of action of the force.

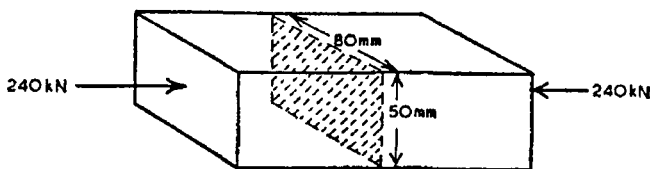


Fig. 139

Fig. 139 shows a direct compressive force of 240 kN applied to a solid piece of material of rectangular cross-section, 80 mm by 50 mm. The cross-sectional area supporting the load is $80 \times 50 \text{ mm} = 4000 \text{ mm}^2$ i.e. $4000 \times 10^{-6} \text{ m}^2$.

$$\begin{aligned}\sigma &= \frac{F}{A} \\ &= \frac{240 \times 10^3}{80 \times 50 \times 10^{-6}} \\ &= 6 \times 10^7 \text{ N/m}^2 \\ &= 60 \text{ MN/m}^2\end{aligned}$$

The material is in compression because the applied force tends to shorten it, and the stress of 60 MN/m^2 is compressive.

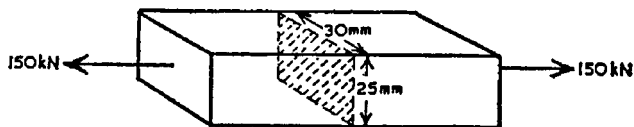


Fig. 140

A direct tensile force of 150 kN carried by a material of cross-section 30 mm by 25 mm is illustrated in Fig. 140.

$$\begin{aligned}\sigma &= \frac{F}{A} \\ &= \frac{150 \times 10^3}{30 \times 25 \times 10^{-6}} = 2 \times 10^8 \text{ N/m}^2 \\ &= 200 \text{ MN/m}^2\end{aligned}$$

The material is in tension because the force tends to stretch it, and the stress of 200 MN/m^2 is tensile.

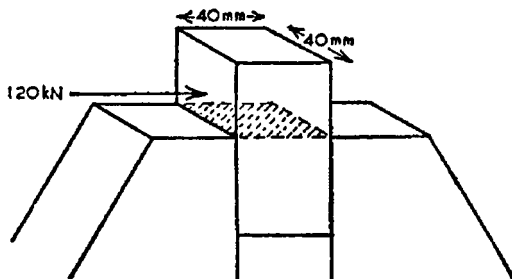


Fig. 141

The next case (Fig. 141) shows a material of rectangular section 40 mm by 40 mm gripped in a vice and a transverse force of 120 kN applied which tends to shear the material.

$$\begin{aligned}\tau &= \frac{F}{A} \\ &= \frac{120 \times 10^3}{40 \times 40 \times 10^{-6}} = 7.5 \times 10^7 \text{ N/m}^2 \\ &= 75 \text{ MN/m}^2\end{aligned}$$

Many cases arise where the material is in 'double shear', that is, two planes of area resist the shearing force. Fig. 142 illustrates a cotter joint and it can be seen that the area of material carrying the force is twice the cross-sectional area of the cotter, therefore the shear stress is Force $\div$ (2 $\times$ cross-sectional area). It is common practice in certain kinds of joints where other factors affect the strength, to assume that the effective area is a little less than twice the cross-section, such as 1.8 or 1.9 times the area.

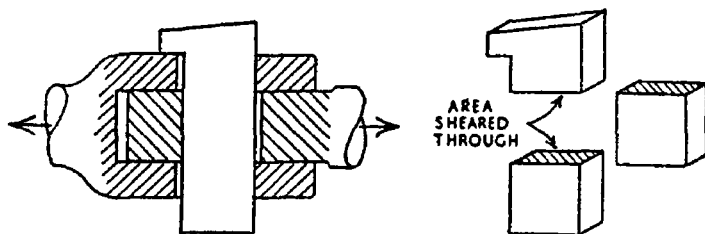


Fig. 142

ULTIMATE TENSILE STRENGTH

The strength of a material is expressed as the stress required to cause fracture. If it is to express the strength in tension, the maximum force required to break the material is divided by the original cross-sectional area at the point of fracture, and is termed the *ultimate tensile strength* (U.T.S), or the *tenacity* of the material. The original area of cross-section is taken as distinct from the actual area at fracture, the latter being much smaller due to the waist or neck which forms just before fracture occurs.

$$\text{Ultimate tensile strength} = \frac{\text{maximum breaking force}}{\text{original area of cross section}}$$

Example. A specimen of mild steel, 20 mm diameter, is tested in a testing machine and snapped when the maximum pull on the specimen was 154 kN. Calculate the tensile strength of this material.

$$\begin{aligned}
 \text{U.T.S} &= \frac{\text{Maximum breaking force}}{\text{original area of cross section}} \\
 &= \frac{154 \times 10^3}{0.7854 \times 0.02^2} \\
 &= 4.901 \times 10^8 \text{ N/m}^2 \\
 &= 490.1 \text{ MN/m}^2 \quad \text{Ans.}
 \end{aligned}$$

Example. A hole 12 mm diameter is to be punched through a plate 18 mm thick. If the shear strength of the material is 300 N/mm², find the load required on the punch.

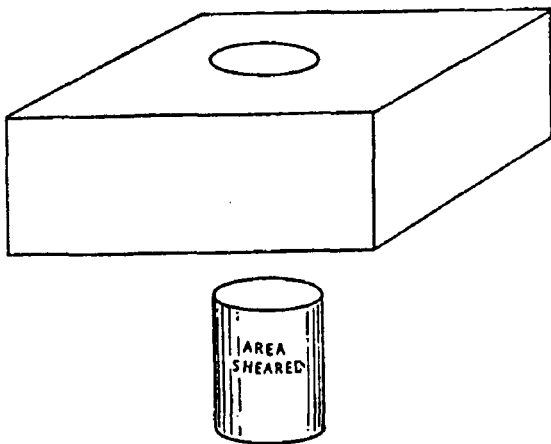


Fig. 143

By reference to Fig. 143 it can be seen that the area of material to be sheared is the circumferential curved surface of the plug to be punched out of the plate. The curved surface area of a cylinder is circumference $\times$ height, therefore:

$$\begin{aligned}
 \text{Area resisting load} &= \text{circumference of plug} \times \text{thickness} \\
 &= \pi \times 12 \times 18 \text{ mm}^2
 \end{aligned}$$

$$\text{Shear strength} = \frac{\text{maximum shear load}}{\text{area resisting load}}$$

$$\therefore \text{Maximum load} = \text{shear strength} \times \text{area}$$

$$\begin{aligned}
 &= 300 \times \pi \times 12 \times 18 \\
 &= 2.036 \times 10^5 \text{ N} \\
 &= 203.6 \text{ kN} \quad \text{Ans.}
 \end{aligned}$$

WORKING STRESS & FACTOR OF SAFETY

It is obvious that the stress allowed in any piece of machinery under working conditions must be much less than that which would cause the metal to fail. A safe working stress is therefore chosen with regard to the conditions under which the material is to work, taking into account whether the stress is due to a dead load, whether it varies or is subject to reversal of stress, and the rapidity of change of stress. For instance, a pressure vessel such as a boiler shell can be regarded as a dead load because the stress in the shell is due to internal pressure which changes very little and therefore a fairly high working stress can be allowed, whereas the piston rod of a double acting reciprocating engine is subjected to rapid varying and reversal stresses and consequently the working stress allowed must be very low.

The ratio of the stress which would cause fracture, to the working stress allowed in the material, is termed the *factor of safety*.

$$\text{Factor of safety} = \frac{\text{breaking stress}}{\text{working stress}}$$

Example. Find the minimum diameter of a mild steel piston rod to carry a maximum load of 560 kN, allowing a factor of safety of 14 and assuming a tensile strength of 420 MN/m².

$$\begin{aligned}
 \text{Working stress} &= \frac{\text{breaking stress}}{\text{factor of safety}} \\
 &= \frac{420}{14} = 30 \text{ MN/m}^2
 \end{aligned}$$

$$\text{Working stress} = \frac{\text{working load}}{\text{area of cross-section}}$$

$$30 \times 10^6 = \frac{560 \times 10^3}{0.7854 \times d^2}$$

$$d = \sqrt{\frac{560 \times 10^3}{30 \times 10^6 \times 0.7854}}$$

$$= 0.01542 \text{ m} = 15.42 \text{ mm} \quad \text{Ans.}$$

STRAIN

Strain is the change of shape that takes place in a material due to it being stressed

LINEAR STRAIN is measured as the change of length per unit length which occurs when a tensile or compressive force is applied thus:

$$\text{Linear strain} = \frac{\text{change of length}}{\text{original length}}$$

$$\epsilon = \frac{x}{l}$$

SHEAR STRAIN is measured by the angular distortion caused by a shear stress. Referring to Fig. 144, which illustrates a material strained by a shear force, the strain is expressed by the angle of distortion which is obtained by dividing x by l . This gives the tangent of the angle ϕ , or, being a small angle, it is also the angle in radians.

$$\text{Shear strain} = \frac{\text{deformation}}{\text{original dimension}}$$

$$\gamma = \frac{x}{y}$$

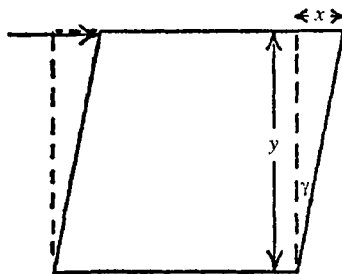


Fig. 144

Change of length and original length are in the same units, therefore in dividing one by the other the units cancel. Hence, strain is a pure number.

ELASTICITY

All metals are elastic to a certain extent. A material will stretch if a tensile force is applied to it, and return to its original length on removal of the force. Similarly, if a compressive force is removed the material will shorten, and when the force is removed the material will return to its original length. Metals are elastic under any form of straining.

There is a limit to this elastic property in every material, known as the *elastic limit*, and if this is exceeded by applying an undue force, the material, after being stretched or compressed, will not return to its original length, but remain permanently set in its deformed condition.

Provided the material is strained within its elastic limit, the amount of strain is directly proportional to the stress. This is known as HOOKE'S LAW. For example, if a piece of metal is subjected to a tensile stress of 75 MN/m^2 which causes a tensile strain of 0.00074, then for a stress of 150 MN/m^2 the strain will be 0.00148, that is, double the stress produces double the strain, and if the stress were trebled the strain would be trebled—always, the strain is proportional to the stress within the limits of elasticity of the metal.

Now consider the following graph plotted during a tensile test on a steel wire 2500 mm long and 0.8 mm diameter (cross-sectional area = 0.5027 mm^2).

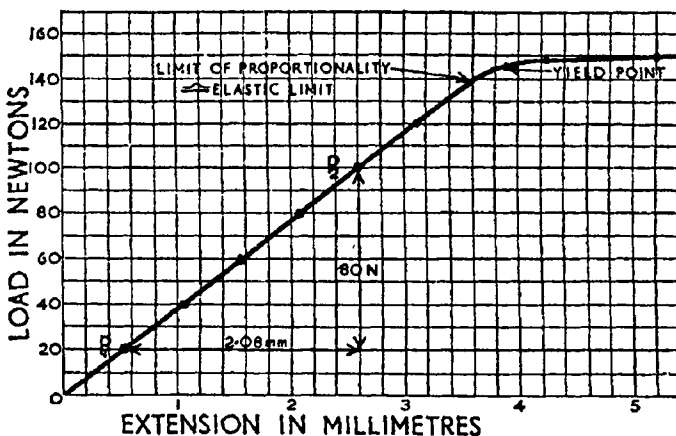


Fig. 145

Part of this graph is a straight line which means that, over this range, extension is proportional to load. As strain depends upon the extension, and stress depends upon the load, it also proves that strain is proportional to stress. If the load were removed at any point over this range, the wire would return to its original length, demonstrating that the material is elastic.

It will be seen that the limit of proportionality, *i.e.* the end of the straight part of the line, is at about the load of 140 N, a little beyond this the material begins to yield and the extension is greatly increased with very little increase of load. The elastic limit is approximately at the limit of proportionality, or in some cases a little higher but before the yield point.

MODULUS OF ELASTICITY

Within the elastic limit, stress is proportional to strain, therefore at any point during the elastic range the value of the stress divided by the corresponding value of the strain will always produce the same result for any one metal. Stress divided by strain is therefore a constant, its value depending upon the material.

When direct (tensile or compressive) stress is divided by direct strain, the constant is termed the *modulus of elasticity* (sometimes Young's modulus), and is represented by E .

$$\text{Modulus of elasticity } \text{N/m}^2 = \frac{\text{direct stress}}{\text{direct strain}}$$

$$E = \frac{\sigma}{\epsilon}$$

When shear stress is divided by shear strain the constant obtained is termed the *modulus of rigidity* and represented by G .

$$G = \frac{\tau}{\gamma}$$

Strain being a pure number, then the units of the moduli of elasticity and rigidity are the same as for stress.

The modulus of elasticity of the wire in the previous example can be obtained from the experimental results. The straight line drawn as near as possible through the plotted points eliminates slight errors of observation during the experiment. Choose any point on this line, divide the value of the stress by the corresponding value of strain, and thus obtain the value of E .

Some graphs however, may not start from zero origin for the strain or extension of the vernier or extensometer is not set to a zero reading at the beginning of the experiment. It is then necessary, and in fact it is always better, to choose two points on the line, such as P_1 and P_2 (see Fig. 145) and divide the increase of stress by the increase of strain, this gives the slope of the graph which is the modulus of elasticity.

Measuring the values from the graph:

$$\text{Increase of load} = 80 \text{ N}$$

$$\begin{aligned} \text{Increase of stress} &= \frac{\text{increase of load}}{\text{area}} \\ &= \frac{80}{0.5027} = 159.2 \text{ N/mm}^2 \end{aligned}$$

$$\text{Increase of extension} = 2.08 \text{ mm}$$

$$\begin{aligned} \text{Increase of strain} &= \frac{\text{increase of extension}}{\text{original length}} \\ &= \frac{2.08}{2500} = 0.000832 \end{aligned}$$

$$\begin{aligned} \text{Modulus of elasticity} &= \frac{\text{stress}}{\text{strain}} \\ &= \frac{\text{increase of stress}}{\text{increase of strain}} \\ &= \frac{159.2}{0.000832} \\ &= 1.914 \times 10^5 \text{ N/mm}^2 \\ E &= 191.4 \text{ kN/mm}^2 \\ &\text{or } 191.4 \text{ GN/m}^2 \end{aligned}$$

TENSILE TEST TO DESTRUCTION

To observe the behaviour of a material subjected to tension up to breaking point, a specimen of the material is machined to a standard size, the ends are gripped in the jaws of a testing machine and a tensile load is slowly applied.

The modulus of elasticity can be obtained as previously described in the wire test, the extensions being read during the elastic range by an extensometer. Before yield point is anticipated the extensometer is removed to prevent damage. The yield point

can be distinctly observed and the extensions from here to breaking point can be measured with little magnification.

The results are plotted on a stress-strain graph and, for mild steel will appear as shown in Fig. 146. When the specimen nears breaking point a waist begins to form in the material and the cross-sectional area becomes smaller. As the waist is forming some of the load can be taken off the machine but the load is removed at a slower rate than that at which the cross-sectional area is reducing, thus the actual stress in the material is still increasing up to breaking point. It is usual, however, to plot the graph of *nominal* stress values which is load divided by the *original* cross-sectional area of the specimen, this accounts for the fall in the graph towards the end near fracture. If the *actual* stress had been plotted it would show the graph continuing to rise. As previously stated, the ultimate tensile strength is taken as the maximum load divided by the original cross-sectional area.

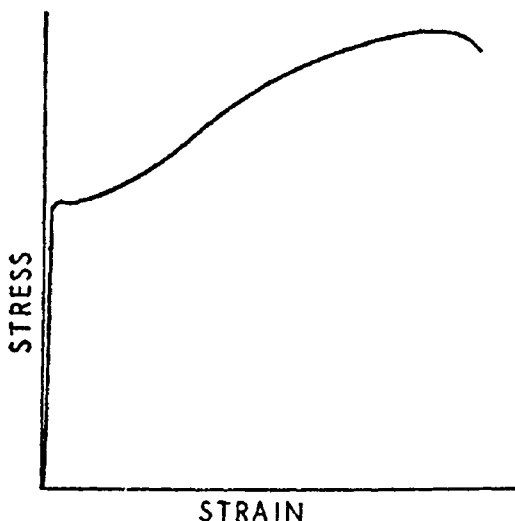


Fig. 146

Further observations can now be made to assist in assessing the quality of the material. The two broken pieces are placed together (see Fig. 147) and the distance between gauge points measured; by subtracting the original gauge length from this measurement, the total elongation is obtained. This is expressed as a percentage of the original length:

$$\begin{aligned}\% \text{ Elongation} &= \frac{\text{total elongation}}{\text{original length}} \times 100 \\ &= \frac{l_2 - l_1}{l_1} \times 100\end{aligned}$$

where l_1 = original gauge length

where l_2 = distance between gauge points after fracture.

By measuring the smallest diameter at the neck of the waist where fracture took place, the final cross-sectional area is obtained and the reduction in cross-sectional area is expressed as a percentage of the original, thus:

$$\begin{aligned}\% \text{ Reduction of area} &= \frac{\text{reduction of cross-sect. area}}{\text{original cross-sect. area}} \times 100 \\ &= \frac{A_1 - A_2}{A_1} \times 100\end{aligned}$$

where A_1 = original cross-sectional area

A_2 = minimum cross-sectional area after fracture.

The percentage elongation and reduction of cross-sectional area are indications of the ductility of the material – that property by which it can be drawn out.

Example. The test piece illustrated in Fig. 147, of diameter 10 mm and gauge length 50 mm, was made from the deposited metal of a welded joint. A tensile test on this specimen produced the following results:

Load at yield point = 25.3 kN

Maximum breaking point = 37.95 kN

Length between gauge points after fracture = 63.5 mm

Diameter at neck of waist after fracture = 7.5 mm

Calculate the stress at yield point, ultimate tensile strength, percentage elongation, percentage contraction of area.

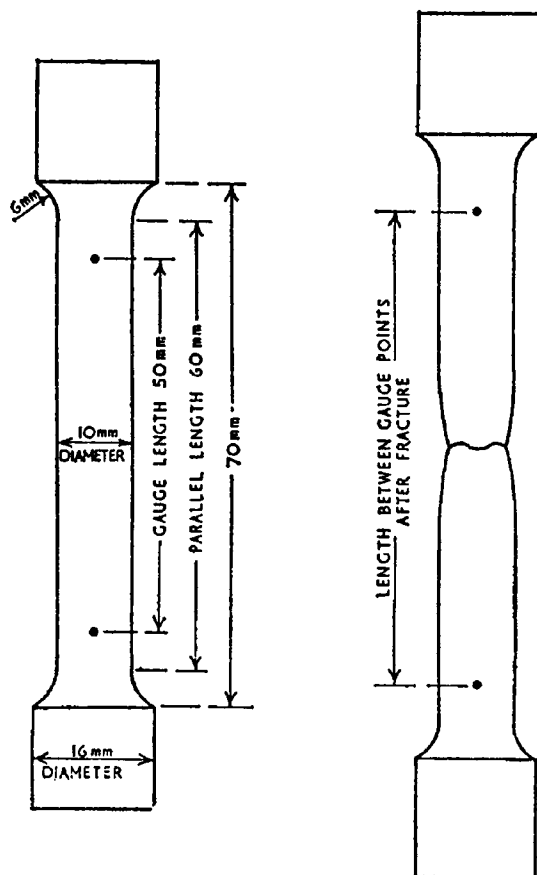


Fig. 147

$$\begin{aligned}\text{Original cross-sectional area} &= 0.7854 \times 10^2 \\ &= 78.54 \text{ mm}^2 = 78.54 \times 10^{-6} \text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{Yield stress} &= \frac{\text{yield load}}{\text{original cross-sect. area}} \\ &= \frac{25.3 \times 10^3}{78.54 \times 10^{-6}} \\ &= 3.221 \times 10^6 \text{ N/m}^2 \\ &= 322.1 \text{ MN/m}^2 \quad \text{Ans. (i)}\end{aligned}$$

$$\begin{aligned}
 \text{U.T.S.} &= \frac{\text{maximum breaking load}}{\text{original cross-sect. area}} \\
 &= \frac{37.95 \times 10^3}{78.54 \times 10^{-6}} = 4.832 \times 10^8 \text{ N/m}^2 \\
 &= 483.2 \text{ MN/m}^2 \quad \text{Ans. (ii)}
 \end{aligned}$$

$$\begin{aligned}
 \% \text{ Elongation} &= \frac{\text{elongation}}{\text{original length}} \times 100 \\
 &= \frac{63.5 - 50}{50} \times 100 = 27 \quad \text{Ans. (iii)}
 \end{aligned}$$

$$\begin{aligned}
 \% \text{ Reduction of area} &= \frac{\text{reduction of cross-sect. area}}{\text{original cross-sect. area}} \times 100 \\
 &= \frac{0.7854 \times 10^2 - 0.7854 \times 7.5^2}{0.7854 \times 10^2} \times 100 \\
 &= \frac{10^2 - 7.5^2}{10^2} \times 100 = 43.75 \quad \text{Ans. (iv)}
 \end{aligned}$$

STRENGTH OF PRESSURE VESSELS

The total force or load due to a fluid pressure is the product of the intensity of the internal pressure and the *projected* area on which the fluid acts. For example, the total load on the piston of an internal combustion engine is the gas pressure multiplied by the projected area of the piston. The projected area of the piston is the same as the cross-sectional area of the bore of the cylinder, this is the effective piston area no matter whether the piston is flat, dished, or has a hump on the top.

Similarly in a pressure vessel, the load causing stress on a section of the shell is the product of the internal pressure and the projected area of the shell perpendicular to that section.

THIN CYLINDERS. A cylindrical vessel whose thickness of shell is small in comparison with the diameter. Boiler drums, air reservoirs and most pipes come into this category.

Consider a thin cylinder without seams, such as a solid drawn pipe or seamless boiler drum, and imagine that the tendency to rupture is by splitting along the longitudinal axis as shown in Fig. 148. The projected area of the curved semi-cylindrical shell is the area of its projected rectangular plan view which is diameter $\times$ length, and the load tending to cause longitudinal rupture is

therefore the intensity of the internal pressure multiplied by this projected area.

Let p = internal fluid pressure,
 d = internal diameter of cylinder,
 l = length of cylinder,
 t = thickness of cylinder,

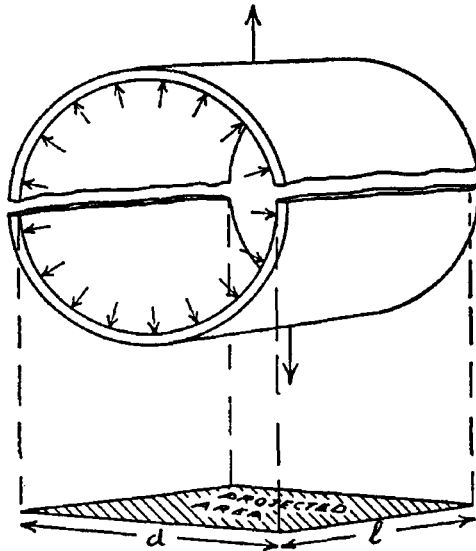


Fig. 148

Load tending to cause longitudinal fracture

$$= \text{internal pressure} \times \text{projected area} \\ = p \times d \times l$$

Area of metal resisting this load

$$= 2 \times \text{thickness} \times \text{length} \\ = 2 \times t \times l$$

Letting the stress along the longitudinal section of the cylinder be represented by σ_1 then:

$$\text{Stress} = \frac{\text{load}}{\text{area}}$$

$$\sigma_1 = \frac{p \times d \times l}{2 \times t \times l}$$

$$\sigma_1 = \frac{pd}{2t} \quad \dots \dots \dots (i)$$

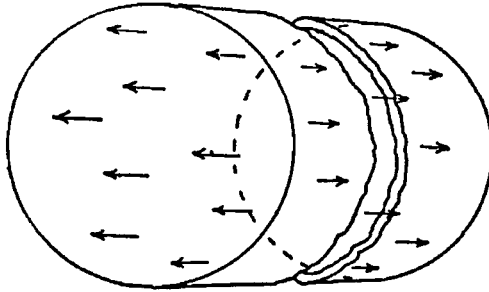


Fig. 149

Now imagine that the tendency is for the cylinder to fracture around its circumference as illustrated in Fig. 149. The cylinder may have flat ends, dished, hemispherical or otherwise, the effective area on which the internal pressure acts is the projected circular area. The total load acting on the circumferential section is therefore the product of the intensity of the internal pressure and the circular area of the end.

The area of metal carrying this load is the area of the annular section $= \frac{\pi}{4}(D^2 - d^2)$. As the thickness is small compared with the diameter, a slight approximation can be made here to simplify the final expression without affecting the calculated result very much by taking the cross-sectional area as circumference multiplied by thickness.

Load tending to cause circumferential fracture

$$= \text{internal pressure} \times \text{projected area}$$

$$= \pi \times \frac{\pi}{4} d^2$$

Area of metal resisting this load.

$$= \text{circumference} \times \text{thickness}$$

$$= \pi d \times t$$

Letting the stress on the circumferential section be represented by σ_2 then:

$$\text{Stress} = \frac{\text{load}}{\text{area}}$$

$$\begin{aligned}\sigma_2 &= \frac{p \times \frac{\pi}{4} d^2}{\pi d \times t} \\ \sigma_2 &= \frac{pd}{4t} \quad \dots \dots \dots \text{(ii)}\end{aligned}$$

From (i) and (ii) the stress in the material is twice as much along the longitudinal section as it is on the circumferential section.

If an elliptical manhole is to be cut in the shell of a cylinder the smallest amount of metal should be cut out of the section which is subjected to the greatest stress, therefore the minor axis should lie in the longitudinal direction and the major axis circumferentially.

Any section of a spherical shell is 'circumferential' therefore expression (ii) only will apply for spherical vessels or spherical ends of cylindrical vessels.

Example. Calculate the stresses on the longitudinal and circumferential sections of a boiler steam drum 2.15 m diameter and 45 mm thick, when working pressure is 30 bar (30×10^5 N/m²). If the tensile strength of the material is 490 MN/m², what factor of safety is allowed?

$$\begin{aligned}\sigma_1 &= \frac{pd}{2t} = \frac{30 \times 10^5 \times 2.15}{2 \times 0.045} \\ &= 7.166 \times 10^7 \text{ N/m}^2 \\ &= 71.66 \text{ MN/m}^2 \quad \text{Ans. (i)}\end{aligned}$$

$$\begin{aligned}\sigma_2 &= \frac{pd}{4t} = \frac{1}{2} \sigma_1 \\ &= 35.83 \text{ MN/m}^2 \quad \text{Ans. (ii)}\end{aligned}$$

$$\begin{aligned}\text{Factor of safety} &= \frac{\text{breaking stress}}{\text{working stress}} \\ &= \frac{490}{71.66} = 6.837 \quad \text{Ans. (iii)}\end{aligned}$$

WORKING PRESSURE. The maximum internal working pressure in a pressure vessel depends upon what is considered to be the maximum allowable stress in the material within the limits of safety. If the vessel is seamless the internal working pressure can

be calculated from the previous expressions, when σ represents the safe working stress, p will represent the safe working pressure.

For a cylindrical vessel the first expression would be used because it involves the greatest stress in the material which is on the longitudinal section, thus:

$$\text{Working pressure} = \frac{2t \times \text{working stress}}{d}$$

For a spherical vessel where all sections are 'circumferential' the working pressure is obtained from:

$$\text{Working pressure} = \frac{4t \times \text{working stress}}{d}$$

SEAM STRENGTH. Cylindrical shells of pressure vessels are often built up of two (sometimes more) curved plates, the seams where these plates are joined together are not as strong as the solid plate and therefore the seam strength must be taken into consideration when determining the safe working pressure.

$$\text{Working pressure} = \frac{2t \times \text{working stress}}{d} \times F_{\text{long}} \quad (\text{iii})$$

$$\text{Working pressure} = \frac{4t \times \text{working stress}}{d} \times F_{\text{circ}} \quad (\text{iv})$$

F represents seam strength fraction.

Consideration of (iii) and (iv) will show that the longitudinal seam should be as near to the strength of the solid plate as practicable, whereas the circumferential seam strength will not affect the safe working pressure unless its strength is less than half the longitudinal seam strength. Longitudinal seams are therefore designed to have a high strength.

Example. Allowing a factor of safety of 5.5, find the safe working pressure in a cylindrical boiler 3.5 mm diameter. The shell plates are mild steel 42 mm thick of tensile strength 460 MN/m², the fractional strength of the longitudinal seams is 0.86 and the fractional strength of the circumferential seams is 0.76.

$$\begin{aligned} \text{Working stress} &= \frac{\text{tensile strength}}{\text{factor of safety}} \\ &= \frac{460 \times 10^6}{5.5} \text{ N/m}^2 \end{aligned}$$

$$\begin{aligned}
 \text{Working pressure} &= \frac{2t \times \text{working stress}}{d} \times F_{\text{long}} \\
 &= \frac{2 \times 0.042 \times 460 \times 10^6 \times 0.86}{3.5 \times 5.5} \\
 &= 1.726 \times 10^6 \text{ N/m}^2 \\
 &= 1.726 \text{ MN/m}^2 \text{ or } 17.26 \text{ bar}
 \end{aligned}$$

With regard to expression (iv) involving the fractional strength of the circumferential seam, as the circumferential seam strength is more than half the longitudinal seam strength, expression (iv) would therefore give a higher working pressure because 4×0.76 is more than 2×0.86 (other quantities being equal) and there is no need to work this out as the lesser value must be taken, hence,

$$\text{Working pressure} = 17.26 \text{ bar} \quad \text{Ans.}$$

STRESSES IN COMPOUND BARS

When two or more bars of equal length carry a load between them the stress in each can be calculated from equations formed, (i) on the principle that, as each is compressed (or stretched) by the same amount, their strains must be equal, and (ii) that the sum of the loads carried by each bar must be equal to the total load carried.

Example. A steel bar of 2580 mm^2 cross-sectional area and a brass bar of 3225 mm^2 cross-sectional area, jointly support a compressive load of 420 kN . Calculate the stress in each bar, and also the strain, taking the moduli of elasticity as:

$$E \text{ for steel} = 207 \text{ GN/m}^2$$

$$E \text{ for brass} = 92 \text{ GN/m}^2$$

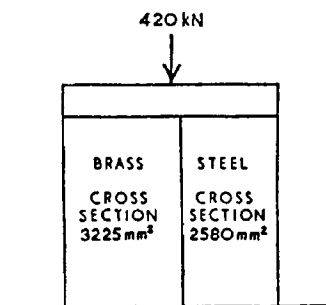


Fig. 150

From $E = \frac{\text{Stress}}{\text{Strain}}$ we have $\text{Strain} = \frac{\text{Stress}}{E}$

and from $\text{Stress} = \frac{\text{Load}}{\text{Area}}$ Load = Stress $\times$ Area

Letting suffix S represent steel and suffix B represent brass:

Strain of steel = Strain of brass

$$\frac{\text{Stress}_S}{E_S} = \frac{\text{Stress}_B}{E_B}$$

$$\begin{aligned}\text{Stress}_S &= \frac{\text{Stress}_B \times E_S}{E_B} \\ &= \frac{\text{Stress}_B \times 207 \times 10^9}{92 \times 10^9}\end{aligned}$$

$$\text{Stress}_S = 2.25 \times \text{Stress}_B \quad \dots \dots \dots \text{(i)}$$

Load carried by steel + Load carried by brass = Total load

$$\text{Stress}_S \times \text{Area}_S + \text{Stress}_B \times \text{Area}_B = 420 \times 10^3$$

$$\text{Stress}_S \times 2580 \times 10^{-6} + \text{Stress}_B \times 3225 \times 10^{-6} = 420 \times 10^3 \quad \text{(ii)}$$

Substituting value of Stress_S from (i) into (ii),

$$2.25 \times \text{Stress}_B \times 2580 \times 10^{-6} + \text{Stress}_B \times 3225 \times 10^{-6} = 420 \times 10^3$$

$$5805 \text{ Stress}_B + 3225 \text{ Stress}_B = 420 \times 10^9$$

$$9030 \text{ Stress}_B = 420 \times 10^9$$

$$\text{Stress}_B = 4.65 \times 10^7$$

$$\sigma_B = 46.5 \text{ MN/m}^2 \quad \text{Ans. (i)}$$

From (i),

$$\text{Stress}_S = 2.25 \times \text{Stress}_B$$

$$\sigma_S = 104.6 \text{ MN.m}^2 \quad \text{Ans. (ii)}$$

The strain is calculated by dividing the stress in either material by its respective modulus of elasticity:

$$\epsilon = \frac{\sigma}{E}$$

$$= \frac{46.5 \times 10^6}{92 \times 10^9} \quad \text{or} \quad \frac{104.6 \times 10^6}{207 \times 10^9}$$

$$= 5.054 \times 10^{-4} \quad \text{Ans. (ii)}$$

EQUIVALENT MODULUS OF ELASTICITY OF COMPOUND BARS

The equivalent modulus of elasticity of compound bars is the value of the modulus of an equivalent single bar which would have the same total area and behave in the same manner as the compound bars under all conditions of loading within the elastic limit.

Let there be two bars, A and B, such as those shown in Fig. 150, of same length, but different cross-sectional areas and different materials.

Let σ = stress, A = area, l = length, x = change of length, E = modulus of elasticity

$$\text{Total load} = \text{load}_A + \text{load}_B = \sigma_A A_A + \sigma_B A_B$$

$$\text{Total area} = A_A + A_B$$

$$E = \frac{\text{stress}}{\text{strain}} = \frac{\text{load}}{\text{area} \times \text{strain}}$$

$$\text{Strain} = \frac{\text{stress}}{E} \text{ and the strain being the same for both bars}$$

$$\text{substitute } \frac{\sigma_A}{E_A} \text{ for strain:}$$

$$\begin{aligned} \text{Equivalent } E &= \frac{\text{total load}}{\text{total area} \times \text{strain}} \\ &= \frac{(\sigma_A A_A + \sigma_B A_B) \times E_A}{(A_A + A_B) \times \sigma_A} \dots \dots (i) \end{aligned}$$

$$\text{strain}_A = \text{strain}_B \quad \therefore \frac{\sigma_A}{E_A} = \frac{\sigma_B}{E_B}$$

$$\sigma_B = \frac{\sigma_A \times E_B}{E_A} \text{ substituting this into (i):}$$

$$\begin{aligned} \text{Equiv. } E &= \frac{\left\{ \sigma_A A_A + \frac{\sigma_A E_B}{E_A} \times A_B \right\} \times E_A}{(A_A + A_B) \times \sigma_A} \\ &= \frac{A_A E_A + A_B E_B}{A_A + A_B} \end{aligned}$$

Taking the compound bar in the previous example,

$$E = \frac{\text{stress}}{\text{strain}} = \frac{\text{load}}{\text{area} \times \text{strain}}$$

Equivalent E of the compound bar *from first principles*,

$$\begin{aligned}
 &= \frac{\text{total load}}{\text{total area}} \times \frac{1}{\text{strain}} \\
 &= \frac{420 \times 10^3}{(3225 + 2580) \times 10^{-6}} \times \frac{1}{5.054 \times 10^{-4}} \\
 &= \frac{420 \times 10^{13}}{5805 \times 5.054} \\
 &= 1.431 \times 10^{11} \text{ N/m}^2 \text{ or } 143.1 \text{ GN/m}^2
 \end{aligned}$$

Alternatively, the expression derived above *could* be used:

$$\begin{aligned}
 \text{Equiv. } E &= \frac{A_s E_s + A_B E_B}{A_s + A_B} \\
 &= \frac{2580 \times 10^{-6} \times 207 \times 10^9 + 3225 \times 10^{-6} \times 92 \times 10^9}{2580 \times 10^{-6} + 3225 \times 10^{-6}} \\
 &= \frac{2580 \times 207 + 3225 \times 92}{2580 + 3225} \times 10^9 \\
 &= \frac{830760 \times 10^9}{5805} \\
 &= 1.431 \times 10^{11} \text{ N/m}^2 \text{ as above.}
 \end{aligned}$$

STRESS DUE TO RESTRICTED THERMAL EXPANSION

Generally speaking, metals expand when heated and contract when cooled, the amount of expansion varies with different metals and depends upon its dimensions and temperature change. The *coefficient of linear expansion* of a metal is the expansion lengthwise for each unit of length for each degree increase in temperature. For instance, the coefficient of linear expansion of copper is about 0.000018 per °C, this means that 1 mm length of copper will expand by 0.000018 mm if heated through 1°C. A length of 10 mm would expand 10 times this amount. If heated through 50°C the increase in length would be 50 times that for 1°C.

Thus, if α = coefficient of linear expansion

l = original length

θ = change of temperature.

then:

$$\text{Change of length} = \alpha l \theta$$

If the natural thermal expansion is restricted in any way, the metal will be strained from its natural length.

Example. A bar of copper 40 mm diameter and 500 mm long is firmly fixed at each end so that it cannot expand. If it is now heated through 50°C of temperature, find the strain in the bar, the stress and the equivalent load. Take the values for copper as, $\alpha = 0.000018$ per °C, $E = 125 \text{ GN/m}^2$.

$$\begin{aligned}\text{Thermal expansion if free} &= \alpha l \theta \\ &= 0.000018 \times 500 \times 50 \\ &= 0.45 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Free lengths should then be} &500 + 0.45 \\ &= 500.45 \text{ mm}\end{aligned}$$

If prevented from expanding the bar is then 0.45 mm shorter than it should be, therefore it is compressed 0.45 mm from its natural length of 500.45 mm,

$$\begin{aligned}\epsilon &= \frac{x}{l} \\ &= \frac{0.45}{500.45}\end{aligned}$$

It is usual however to make a slight approximation by taking the original cold length (500 mm) instead of the heated length (500.45). The difference in the final answer is negligible and it is much more convenient for the solution of more complicated problems to follow. It will also be seen that by making this approximation the length of the material is not required to be known because the strain can be obtained direct from:

$$\begin{aligned}\epsilon &= \frac{x}{l} \\ &= \frac{\alpha l \theta}{l} \\ &= \alpha \theta\end{aligned}$$

Hence,

$$\begin{aligned}\epsilon &= \frac{0.45}{500} \text{ or } 0.000018 \times 50 \\ &= 0.0009 \quad \text{Ans. (i)}\end{aligned}$$

$$\begin{aligned}\sigma &= \epsilon \times E \\ &= 0.0009 \times 125 \times 10^9 \\ &= 1.125 \times 10^8 \text{ N/m}^2 \\ &= 112.5 \text{ MN/m}^2 \quad \text{Ans. (ii)}\end{aligned}$$

$$\text{Cross-sectional area of bar} = 0.7854 \times 40^2 \times 10^{-6} \text{ m}^2.$$

$$\sigma = \frac{F}{A}$$

$$\text{Equivalent load} = \text{stress} \times \text{area}$$

$$= 112.5 \times 10^6 \times 0.7854 \times 40^2 \times 10^{-6}$$

$$= 1.414 \times 10^5 \text{ N}$$

$$= 141.4 \text{ kN}$$

Ans. (iii)

f STRESS DUE TO THERMAL EXPANSION OF COMPOUND BARS

If two bars of metal of different coefficients of linear expansion are firmly fastened together and then heated, the bar with the larger coefficient will be prevented from expanding its full amount by the bar with the smallest coefficient, while the latter will be pulled further out from its natural expansion by the former. The bars will therefore be strained.

Consider two bars, A and B. Bar A has a large coefficient of linear expansion (α_A) and bar B has a small coefficient (α_B). If allowed to expand quite freely when heated, the expansion of A will be α_A/θ and the expansion of B will be α_B/θ . If these two bars were joined together when cold so that one is prevented from expanding more than the other when they are heated, neglecting any bending tendency, A will exert an outward pull in trying to expand its full amount, and B will exert an inward pull as a resistance to being pulled out more than its natural heated length. The resulting increase of length of the combined bars will be somewhere between the two amounts of free expansion. If the actual increase of length of the bars is represented by y , then:

Bar A is pulled back a distance of $\alpha_A/\theta - y$

Bar B is pulled out by a distance of $y - \alpha_B/\theta$

It can be seen by reference to Fig. 151, which is a magnified illustration, that the sum of the above two quantities is equal to the difference between their free expansions, thus:

$$(\alpha_A/\theta - y) + (y - \alpha_B/\theta) = \alpha_A/\theta - \alpha_B/\theta$$

As strain is change of length divided by original length, divide every term by l to obtain an expression in terms of the strain in each bar:

$$\frac{\alpha_A l \theta - y}{l} + \frac{y - \alpha_B l \theta}{l} = \frac{\alpha_A l \theta}{l} - \frac{\alpha_B l \theta}{l}$$

which is:

$$\text{Strain in bar A} + \text{Strain in bar B} = \alpha_A \theta - \alpha_B \theta$$

In words this is:

Sum of the strains = Difference in free expansion per unit length

This equation, coupled with the fundamental fact that for equilibrium:

Outward pull of one bar = Inward pull of the other provides the basis of solution for problems on this subject.

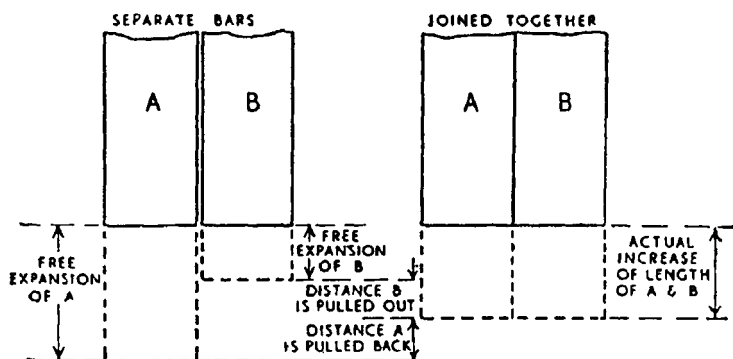


Fig. 151

Example. A solid bar of steel 80 mm diameter is placed inside a brass tube 80 mm inside diameter, 10 mm thick and of equal length. The two ends of the bar and tube are firmly fixed together and the whole is heated through 100°C . Find the stress in the steel and brass taking the values:

Coeff. of linear exp. for steel = 11×10^{-6} per $^\circ\text{C}$

Coeff. of linear exp. for brass = 19×10^{-6} per $^\circ\text{C}$

Modulus of elasticity for steel = 206 GN/m^2

Modulus of elasticity for brass = 103 GN/m^2

The brass tends to expand more than the steel because of its higher coefficient of linear expansion.

Outward pull of the brass = Inward pull of the steel

$\text{STRESS}_{\text{BRASS}} \times \text{Area}_{\text{BRASS}} = \text{STRESS}_{\text{STEEL}} \times \text{Area}_{\text{STEEL}}$

$$\text{Stress}_B \times 0.7854 (100^2 - 80^2) = \text{Stress}_S \times 0.7854 \times 80^2$$

$$\text{Stress}_B \times 3600 = \text{Stress}_S \times 6400$$

$$\text{Stress}_B = \text{Stress}_S \times \frac{16}{9} \dots \dots (i)$$

Sum of the strains = Difference in expansion per unit length.

$$\frac{\text{Stress}_B}{E_B} + \frac{\text{Stress}_S}{E_S} = \alpha_B \theta - \alpha_S \theta$$

Substituting equivalent of Stress_B from (i) and inserting values of E , α and θ ,

$$\frac{\text{Stress}_S \times 16}{103 \times 10^9 \times 9} + \frac{\text{Stress}_S}{206 \times 10^9} = 100 \times 10^{-6} (19 - 11)$$

Multiplying throughout by $206 \times 10^9 \times 9$ and simplifying:

$$32 \text{ Stress}_S + 9 \text{ Stress}_S = 100 \times 10^{-6} \times 8 \times 206 \times 10^9 \times 9$$

$$41 \text{ Stress}_S = 8 \times 206 \times 9 \times 10^5$$

$$\sigma_S = 3.618 \times 10^7 \text{ N/m}^2$$

$$= 36.18 \text{ MN/m}^2 \quad \text{Ans. (i)}$$

From (i),

$$\sigma_B = 36.18 \times \frac{16}{9}$$

$$= 64.32 \text{ MN/m}^2 \quad \text{Ans. (ii)}$$

f ELASTIC STRAIN ENERGY

The work done by a force is the product of the average force applied and the distance through which the force moves.

Consider compressing a helical spring, let this spring have a stiffness of say, 100 N per mm, that is, it will compress (or stretch) 1 mm if an axial load of 100N is placed on it, or 2 mm if it carries 200 N, etc. Imagine a load of 400 N being lowered by a crane gradually on to the spring. Whilst lowering, at the moment when the spring is compressed 1 mm, the spring is carrying 100 N of the load and the crane is still holding the remaining 300 N. At the moment when the spring is compressed 2 mm, the spring is then carrying 200 N of the load and the crane is carrying the other 200 N. At 3 mm of compression the spring now carries 300 N of the load and the crane only the remaining 100 N. At 4 mm of compression, the whole load of 400 N rests on the spring and the

crane carries none, the crane hook can now be uncoupled. Thus the load on the spring was gradually increased from zero to the full 400 N and the *average* load applied to the spring during its compression up to 4 m is:

$$\text{Average load} = \frac{1}{2}(0 + 400) = 200 \text{ N}$$

The work done in compressing the spring is the average load multiplied by the distance it moves, thus:

$$\begin{aligned}\text{Work done} &= \text{Average load} \times \text{compression} \\ &= 200 \times 4 \\ &= 800 \text{ N mm} = 0.8 \text{ N m} = 0.8 \text{ J}\end{aligned}$$

This is one method of storing energy, by straining the material within its elastic limit. If a spring is compressed and locked in that position by a trigger, it will contain potential energy by virtue of its strained condition, on releasing the trigger the spring can do work as it returns to its original length, the work given out equal to the work which was put into the spring to compress it.

The above is a magnification of what happens when a load is gradually applied on a bar of metal (all loads are taken as being gradually applied unless stated to be otherwise). In metals the amount of compression (or extension) is very small in comparison with a spring but the principle is the same providing the metal is strained within the elastic limit.

The energy stored in a material strained within the elastic limit is termed the *elastic strain energy* or *resilience* and this quantity is equal to the work done to strain the material.

Let W = final load on the material

„ x = compression or extension

$$\begin{aligned}\text{Work done to strain material} &= \text{Average load} \times \text{distance} \\ &= \frac{1}{2}(0 + W) \times x \\ &= \frac{1}{2}Wx\end{aligned}$$

Fig. 152 is a graph showing the relationship between load and extension or compression when loaded within the elastic range, and it can be seen that the area under the graph represents work done (see also Chapter 4).

For convenience the load can be expressed in terms of the stress in the material, its dimensions, and its modulus of elasticity, this is more useful because it expresses the elastic strain energy with regard to its strength properties:

$$\begin{aligned}\text{Load} &= \text{stress} \times \text{cross-sect. area} \\ x &= \text{strain} \times \text{length} \\ &= \frac{\text{stress}}{E} \times \text{length}\end{aligned}$$

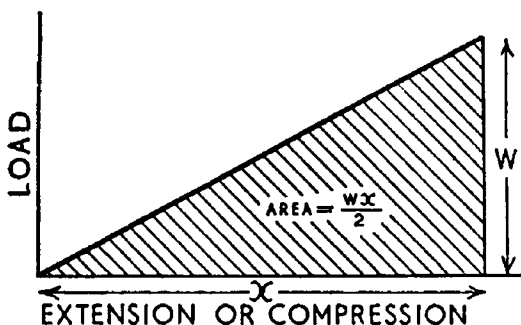


Fig. 152

Substituting these values of load (W) and extension (x)

$$\begin{aligned}
 \text{Elastic strain energy} &= \text{work done} \\
 &= \frac{1}{2} Wx \\
 &= \frac{\text{stress} \times \text{area} \times \text{stress} \times \text{length}}{2 \times E}
 \end{aligned}$$

Cross-sectional area $\times$ length = volume, therefore,

$$\text{Elastic strain energy} = \frac{\text{stress}^2 \times \text{volume}}{2E} = \frac{\sigma^2 V}{2E}$$

Example. Calculate the elastic strain stored in a solid steel bar 75 mm diameter and 2 m long when subjected to a direct tensile stress of 200 MN/m^2 , taking the modulus of elasticity of this material as 207 GN/m^2 .

$$\begin{aligned}
 \text{Elastic strain energy} &= \frac{\sigma^2 V}{2E} \\
 &= \frac{(200 \times 10^6)^2 \times 0.7854 \times 0.075^2 \times 2}{2 \times 207 \times 10^9} \text{ Nm} \\
 &= 853.7 \text{ J} \qquad \text{Ans.}
 \end{aligned}$$

Example. Compare the resilience of two solid round bars, A and B, of similar material, equal length, and carrying an equal direct compressive load; bar A has a constant diameter over its entire length, bar B has the same diameter as A for three-quarters of its length and reduced to half diameter for its remaining quarter-length.

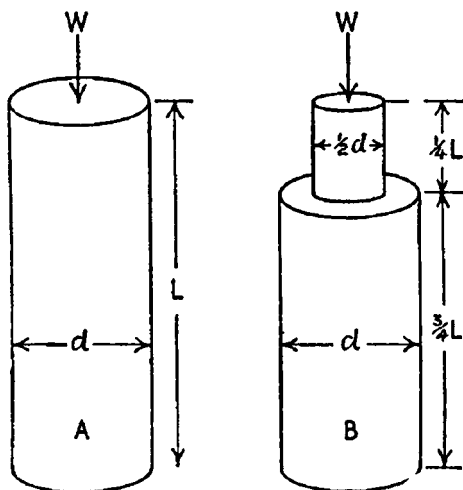


Fig. 153

Referring to Fig. 153.

Resilience of A : Resilience of B

$$\frac{\text{stress}^2 \times \text{vol.}}{2E} : \frac{\text{stress}^2 \times \text{vol.}}{2E} \text{ of large part} +$$

$$\frac{\text{stress}^2 \times \text{vol.}}{2E} \text{ of small part}$$

$$\text{volume} = \text{area} \times \text{length}$$

$$\text{and, stress} = W \div \text{area}$$

$$\frac{\text{Stress}^2 \times \text{volume}}{2E} \text{ can be written:}$$

$$\frac{W^2 \times \text{area} \times \text{length}}{\text{area}^2 \times 2E} = \frac{W^2 \times \text{length}}{\text{area} \times 2E}$$

$$\text{Also, area of diameter } d = 0.7854d^2$$

$$\begin{aligned} \text{,,} \quad \text{,,} \quad \frac{1}{2}d &= 0.7854 \left(\frac{1}{2}d\right)^2 \\ &= 0.7854 \times \frac{1}{4}d^2 \end{aligned}$$

And, $2E$ cancels from every term, therefore Resilience ratio A to B becomes,

$$\frac{W^2 \times L}{0.7854 d^2} : \frac{W^2 \times \frac{3}{4}L}{0.7854 d^2} + \frac{W^2 \times \frac{1}{4}L}{0.7854 \times \frac{1}{4}d^2}$$

W^2 , L , 0.7854 , and d^2 cancels from every term:

$$1 : \frac{3}{4} + 1$$

$$1 : 1\frac{3}{4} \text{ Ans.}$$

f SUDDENLY APPLIED AND SHOCK LOADS

Reverting back to the consideration of a load applied on a spring, this load was gradually applied. If, instead, the same load was held just about touching the top coil and suddenly released, the effect would be to compress the spring initially to twice the normal amount of compression, that is 8 mm instead of 4 mm, the spring and load would then vibrate up and down until, dampened by friction, the vibrations would die out and the spring settle down with its normal compression of 4 mm.

The important point to note is that the *initial* compression due to this *suddenly applied load* is twice as much as it would be had the same load been gradually applied, the strain in the material is doubled and therefore the stress is doubled.

Further, if the same load were allowed to fall on to the spring from a height h above the top coil, the impact of this shock load would cause a much greater initial strain and stress in the material, the amount depending upon the height h from which the load was allowed to fall before striking the spring.

Now consider a solid bar of metal, let a load W fall through a distance h before striking the bar, and let the initial amount of compression be x . Provided the initial stress is within the elastic limit, then:

Potential Energy given up = Elastic Strain Energy
by falling weight taken up by bar

$$W \times (h + x) = \frac{\sigma^2 V}{2E}$$

from which the stress can be calculated if the other quantities are known.

If the value of h is nil, it becomes the case of a 'suddenly applied load' without impact, thus:

$$W \times x = \frac{\sigma^2 V}{2E}$$

$$\begin{aligned} \text{Substituting, } x &= \text{strain} \times \text{length} \\ &= \frac{\text{stress}}{E} \times \text{length} \end{aligned}$$

and volume = area $\times$ length:

$$W \times \frac{\text{stress}}{E} \times \text{length} = \frac{\text{stress}^2 \times \text{area} \times \text{length}}{2E}$$

$$W = \frac{\text{stress} \times \text{area}}{2}$$

$$\sigma = \frac{2W}{A}$$

This shows that the initial stress caused by a suddenly applied load is twice the stress that would be produced if the same load had been gradually applied.

f Example. A load of 10 kN falls 6 mm on to the end of a vertical bar 200 mm long and 2500 mm² cross-sectional area. Calculate the instantaneous compression and stress, taking the modulus of elasticity of the material as 195 GN/m². If the same load had been (*b*) gradually applied, (*c*) suddenly applied without impact, what would be the stress in the bar?

$$\begin{aligned} \text{Potential Energy lost} &= \text{Elastic Strain energy} \\ \text{by falling weight} &\quad \text{taken up by bar} \\ W \times (h + x) &= \frac{\sigma^2 V}{2E} \end{aligned}$$

There are two unknowns in this case, therefore one must be expressed in terms of the other.

Stress in terms of x :

$$\begin{aligned} \text{Stress} &= \text{strain} \times E \\ &= \frac{x}{\text{length}} \times E \end{aligned}$$

Substituting for stress and also area $\times$ length for volume:

$$\begin{aligned} W(h + x) &= \frac{x^2 \times E^2 \times \text{area} \times \text{length}}{\text{length}^2 \times 2 \times E} \\ h + x &= \frac{x^2 \times E \times \text{area}}{\text{length} \times 2 \times W} \\ 0.006 + x &= \frac{x^2 \times 195 \times 10^9 \times 2500 \times 10^{-6}}{0.2 \times 2 \times 10 \times 10^3} \\ 0.006 + x &= 1.281 \times 10^5 \times x^2 \end{aligned}$$

$$1.218 \times 10^5 x^2 - x - 0.006 = 0$$

Solving this quadratic,

$$\begin{aligned} x &= \text{instantaneous compression} \\ &= 0.2261 \times 10^{-3} \text{ m} \\ &= 0.2261 \text{ mm} \quad \text{Ans. a(i)} \end{aligned}$$

$$\begin{aligned} \text{Stress} &= \text{strain} \times E \\ &= \frac{x}{\text{length}} \times E \\ &= \frac{0.2261 \times 195 \times 10^9}{200} \\ &= 2.205 \times 10^8 \text{ N/m}^2 \\ &= 220.5 \text{ MN/m}^2 \quad \text{Ans. a(ii)} \end{aligned}$$

Note that x is a small quantity and makes only a small difference when added to h . An approximate value of the stress can be obtained by neglecting x thus:

$$W \times (h + x) = \frac{\sigma^2 V}{2E}$$

Neglecting x ,

$$W \times h = \frac{\sigma^2 V}{2E}$$

$$\text{Stress} = \sqrt{\frac{2EWh}{V}}$$

If the same load of 10 kN had been gradually applied:

$$\begin{aligned}\sigma &= \frac{W}{A} \\ &= \frac{10 \times 10^3}{2500 \times 10^{-6}} \\ &= 4 \times 10^6 \text{ N/m}^2 \\ &= 4 \text{ MN/m}^2 \quad \text{Ans. (b)}\end{aligned}$$

If the same load had been suddenly applied:

$$\begin{aligned}\sigma &= \frac{2W}{A} \\ &= 8 \text{ MN/m}^2 \quad \text{Ans. (c)}\end{aligned}$$

f STRESSES ON OBLIQUE PLANES

Fig. 154 shows a tensile load P_1 applied to a material, if the area of the plane normal (at right angles or perpendicular) to the direction of the load is represented by a_1 and the stress on this plane by σ_1 then the total pull, being stress $\times$ area, can be represented by $\sigma_1 \times a_1$.

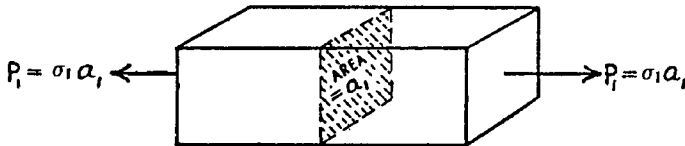


Fig. 154

Consider now a plane in the material inclined at θ° to the normal cross-section (Fig. 155), the effect of the pull P_1 in the

direction of the axis is to produce two loads on the oblique plane by virtue of the rectangular components of P_1 , one of these being perpendicular to the oblique plane and the other tangential to it.

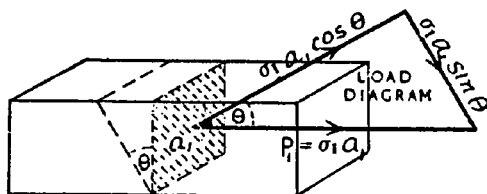


Fig. 155

Referring to the load diagram, the load perpendicular to the oblique plane is $\sigma_1 a_1 \cos \theta$ and the tangential load is $\sigma_1 a_1 \sin \theta$.

The area of the oblique plane is $a_1 \div \cos \theta$ and, stress being load $\div$ area, the normal tensile stress σ_n on this oblique plane is,

$$\sigma_n = \sigma_1 a_1 \cos \theta \div \frac{a_1}{\cos \theta}$$

$$\sigma_n = \sigma_1 \cos^2 \theta \quad \dots \dots \dots \text{(i)}$$

The tangential load produces a shear stress τ thus:

$$\tau = \sigma_1 a_1 \sin \theta \div \frac{a_1}{\cos \theta}$$

$$\tau = \sigma_1 \sin \theta \cos \theta$$

or, since $\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$, this may be written:

$$\tau = \frac{1}{2} \sigma_1 \sin 2\theta \quad \dots \dots \dots \text{(ii)}$$

The maximum value of τ will be when the value of $\sin 2\theta$ is a maximum. The maximum sine is unity which is for an angle of 90° . If 2θ is 90° then $\theta = 45^\circ$, therefore the shear stress will be maximum across a plane inclined at 45° .

$$\tau_{\max} = \frac{1}{2} \sigma_1 \sin (2 \times 45^\circ)$$

$$= \frac{1}{2} \sigma_1$$

Example. A bar 25 mm diameter is subjected to a direct tensile force of 60 kN. Calculate the normal and shear stresses on a plane at 35° to the flat end of the bar. Calculate also the maximum shear stress in the material.

$$\sigma_1 = \frac{W}{A} = \frac{60 \times 10^3}{0.7854 \times 25^2 \times 10^{-6}}$$

$$= 1.222 \times 10^8 \text{ N/m}^2 = 122.2 \text{ MN/m}^2$$

$$\begin{aligned}\sigma_n &= \sigma_1 \cos^2 \theta \\ &= 122.2 \times 0.8192^2 \\ &= 82.04 \text{ MN/m}^2 \quad \text{Ans. (i)}\end{aligned}$$

$$\begin{aligned}\tau &= \frac{1}{2} \sigma_1 \sin 2\theta \\ &= \frac{1}{2} \times 122.2 \times 0.9397 \\ &= 57.41 \text{ MN/m}^2 \quad \text{Ans. (ii)}\end{aligned}$$

$$\begin{aligned}\tau_{\max} &= \frac{1}{2} \sigma_1 \\ &= 61.1 \text{ MN/m}^2 \quad \text{Ans. (iii)}\end{aligned}$$

If the bar in Fig. 155 is subjected to a tensile load P_2 only, in a direction at right angles to the axis, as shown in Fig. 156 then, considering the equilibrium of a piece of the material such as the wedge shaped portion, let a_2 be the area normal to the load P_2 and the stress in this direction be represented by σ_2 , the load can therefore be represented by $\sigma_2 a_2$.

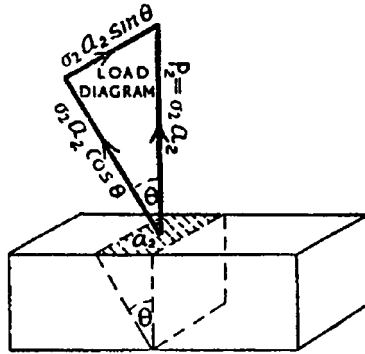


Fig. 156

The rectangular components of this load, normal and tangential to the oblique plane are $\sigma_2 a_2 \sin \theta$ and $\sigma_2 a_2 \cos \theta$. The area of the oblique plane is $a_2 \div \sin \theta$.

The normal tensile stress on the oblique plane σ_n due to P_2 is therefore:

$$\begin{aligned}\sigma_n &= \sigma_2 a_2 \sin \theta \div \frac{a_2}{\sin \theta} \\ \sigma_n &= \sigma_2 \sin^2 \theta \quad \dots \dots \dots \text{(iii)}\end{aligned}$$

The shear stress on the oblique plane due to P_2 is:

$$\tau = \sigma_2 a_2 \cos \theta \div \frac{a_2}{\sin \theta}$$

$$\tau = \sigma_2 \sin \theta \cos \theta$$

$$\tau = \frac{1}{2} \sigma_2 \sin 2\theta \quad \dots \dots \dots \quad \text{(iv)}$$

Combining the above results when tensile stresses of σ_1 and σ_2 at right angles to each other act at the same time on a material, the normal stress σ_n and the shear stress τ on the oblique plane are the algebraic sums of the effects of σ_1 and σ_2 as follows, and illustrated in Fig. 157.

From (i) and (iii),

$$\sigma_n = \sigma_1 \cos^2 \theta + \sigma_2 \sin^2 \theta \quad \dots \dots \dots \quad \text{(v)}$$

From (ii) and (iv),

$$\tau = \frac{1}{2} \sigma_1 \sin 2\theta - \frac{1}{2} \sigma_2 \sin 2\theta$$

$$\tau = \frac{1}{2} \sin 2\theta (\sigma_1 - \sigma_2) \quad \dots \dots \dots \quad \text{(vi)}$$

Note that (iv) was subtracted from (ii) to obtain (vi) because the shear stresses are opposite in direction.

The above expressions have been derived on the assumption that σ_1 and σ_2 are *tensile* stresses. If one of these or both were compression, they may be regarded as negative tensile stresses and the above will hold good if the appropriate signs are used.

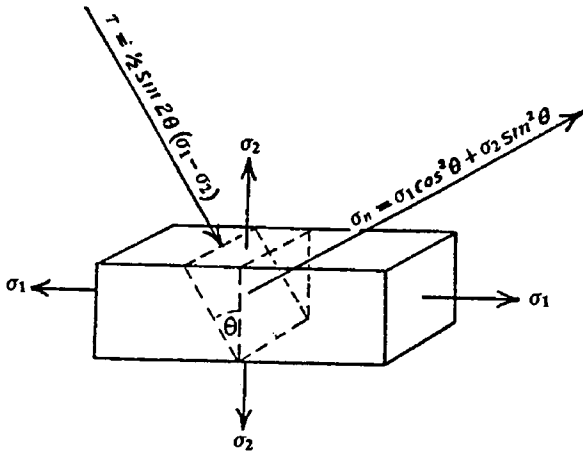


Fig. 157

Example. The circumferential tensile stress in the material of the shell plates of a starting air reservoir, due to the internal pressure, is 68 MN/m^2 , and the longitudinal tensile stress is 34 MN/m^2 . Calculate the normal and shear stresses on a plane of the material at 45° to the axis.

$$\begin{aligned}\sigma_n &= \sigma_1 \cos^2 45^\circ + \sigma_2 \sin^2 45^\circ \\ &= 68 \times 0.7071^2 + 34 \times 0.7071^2 \\ &= 68 \times 0.5 + 34 \times 0.5 \\ &= 51 \text{ MN/m}^2 \quad \text{Ans. (i)}\end{aligned}$$

$$\begin{aligned}\tau &= \frac{1}{2} \sin 2\theta (\sigma_1 - \sigma_2) \\ &= \frac{1}{2} \sin 90^\circ (68 - 34) \\ &= \frac{1}{2} \times 1 \times 34 \\ &= 17 \text{ MN/m}^2 \quad \text{Ans. (ii)}\end{aligned}$$

COMPLEMENTARY SHEAR STRESS. If a rectangular element shown in Fig. 158, of thickness t , is subject to an applied shear stress τ there is a couple $\tau(xt)y$ acting on the element. For equilibrium this must be balanced by a couple τ^1 .

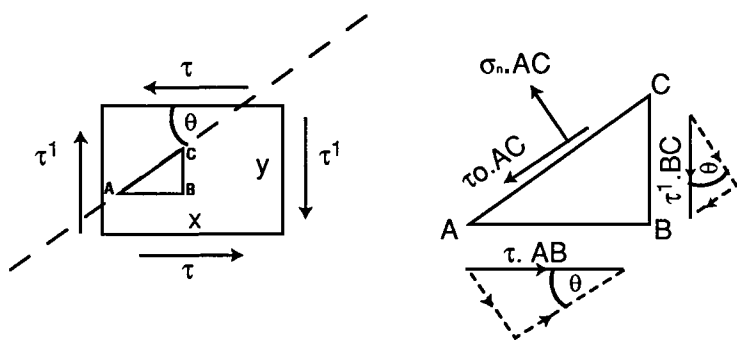


Fig. 158

For equilibrium:

$$\begin{aligned}\tau(xt)y &= \tau^1(yt)x \\ \tau &= \tau^1\end{aligned}$$

Complementary shear stress τ^1 is equal in magnitude to the applied shear stress τ (both stresses act either towards or away from a corner).

Consider the triangular prism of material ABC and resolving forces perpendicular and parallel to the plane AC:

$$\begin{aligned}
 \sigma_n ACt &= \tau^1 BCt \cos \theta + \tau ABt \sin \theta \\
 \sigma_n &= \tau^1 \sin \theta \cos \theta + \tau \cos \theta \sin \theta \\
 \sigma_n &= 2\tau \sin \theta \cos \theta \\
 \sigma_n &= \tau \sin 2\theta \quad \text{normal, tensile} \\
 \tau_\theta ACt &= \tau ABt \cos \theta - \tau^1 BCt \sin \theta \\
 \tau_\theta &= \tau \cos^2 \theta - \tau^1 \sin^2 \theta \\
 \tau_\theta &= \tau \cos 2\theta \quad \text{shear}
 \end{aligned}$$

TEST EXAMPLES 9

1. A tie bar made of mild steel of tensile strength 462 MN/m^2 is to carry a tensile load of 11.12 kN , find its diameter allowing a factor of safety of 12.

2. Calculate the safe load in kN that can be carried by a stud of 580.2 mm^2 cross-sectional area at the bottom of the thread allowing a safe working stress of 35 MN/m^2 . Calculate also the number of these studs required to hold the cylinder cover of a diesel engine where the maximum pressure in the cylinder is 42 bar ($42 \times 10^5 \text{ N/m}^2$) and the diameter of the cover inside the joint is 380 mm .

3. In a cotter joint such as is shown in Fig. 142, the cotter is 100 mm wide and 25 mm thick. The tensile strength of the stay is 440 MN/m^2 and the shear strength of the cotter is 385 MN/m^2 . Allowing 1.8 times the cross-sectional area of the cotter as being effective instead of twice the area for double shear, calculate the diameter of the stay and the diameter of its swelled plug end which is slotted to take the cotter, so that all parts are equal strength.

4. A tensile test is performed on a length of wire of 2 mm diameter and 4 m long, and the following data observed:

Load (N)	100	200	300	400	500	600
Extension (mm)	0.6	1.2	1.8	2.4	3.0	3.6

Plot a load-extension graph and find the modulus of elasticity of the material.

5. The following data were taken during a tensile test on a specimen of mild steel 20 mm diameter and 100 mm long between gauge points:

Yield load = 85.4 kN

Breaking load = 143.6 kN

Diameter at waist after fracture = 14.8 mm

Length between gauge points after fracture = 127 mm

Calculate (i) stress at yield point, (ii) ultimate tensile strength, (iii) percentage elongation, (iv) percentage reduction of area.

6. A cylindrical boiler, working at a pressure of 10 bar, is to have an internal diameter of 3 m. The maximum permissible tensile stress in the shell plate is 80 MN/m^2 .

Calculate the minimum plate thickness (a) if joints are riveted, having a joint efficiency of 70%, (b) if the joints are welded.

The welded joints can be assumed to have the same strength as the parent metal of the shell plate.

7. Three vertical wires, each 4.5 m long, hang in the same vertical plane from a horizontal ceiling and carry a mass of 203.9 kg between them at their lower ends. The two outside wires are of steel and are each 4 mm diameter, the middle wire is brass 5 mm diameter. Find (i) the stress in each wire, (ii) the load carried by each wire, (iii) the stretch. E for steel = 200 GN/m^2 , E for brass = 100 GN/m^2 .

8. A straight steel steam pipe is to be fitted between two rigid bulkheads 1.5 m apart. When heated due to steam passing through, the compressive stress set up in the pipe must not exceed 177 MN/m^2 . If the free expansion of the pipe in the heated condition would be 2 mm and the modulus of elasticity of the steel is 210 GN/m^2 , calculate the tensile stress to induce in the pipe on assembly when cold.

9. A bronze bar 50 mm diameter and 1.25 m long is subjected to a tensile stress of 75 MN/m^2 . Find the extension and the work done to stretch the bar. Take E for bronze = 85 GN/m^2 .

f10. A bar of steel is covered by a copper sheath over its entire length and the sheath is firmly fixed to the bar so that one cannot expand more than the other. The cross-sectional area of the copper sheath is half the cross-sectional area of the steel bar. Find the stresses in the steel and the copper when this compound bar is heated through 100°C , taking the values:

Coeff. of linear expansion for steel = 12×10^{-6} per $^{\circ}\text{C}$

Coeff. of linear expansion for copper = 17×10^{-6} per $^{\circ}\text{C}$

E for steel = 206 GN/m^2

E for copper = 103 GN/m^2

f11. A bar of hexagonal cross-section measures 50 mm across the hexagonal face from one corner to its opposite corner. Find the percentage increase in resilience that would result by drilling a hole 30 mm diameter axially through the centre of the bar from end to end.

f12. A mass of 102 kg falls through 12 mm directly on to the end of a bar 25 mm diameter and 1.5 m long. Taking the modulus of elasticity of the material as 207 GN/m^2 , find the instantaneous initial stress produced.

f13. A piston of 2 t mass is suspended from an engine room crane at the end of a steel cable 15 mm dia. ($E = 210 \text{ GN/m}^2$). Calculate the stress in the cable. When the piston is being lowered at a steady speed of 0.2 m/s and is 10 m below the crane, the crane drum jams and stops suddenly. Calculate:

(a) the instantaneous extension of the cable caused by the stoppage.

(b) the momentary additional stress in the cable due to the sudden stop.

f14. A bar 20 mm diameter is subjected to a direct pull of 50 kN. Find the normal and shear stresses on a plane at 45° to the axis.

f15. Explain the meaning of the term complementary shear stress. A horizontal beam of solid rectangular section is subject at one point to a pure shear stress of 15 MN/m^2 acting in a vertical direction. Determine the normal and tangential stresses on a plane in the beam at 60° to the horizontal.